Smart Healthcare:
Malaria detection using CNN models
CS 5262 - Final Project Report

Arpit Ojha [1] **Shruti Jain** [1]

Abstract

Malaria continues to pose significant health risks globally, especially in sub-Saharan Africa. For diagnosis of malaria, a technician has to look into the blood smear slides for the presence of the Plasmodium parasites within the red blood cells. Conventional microscopic diagnosis has challenges, such as a lack of trained personnel, adequate laboratory facilities, and quality assurance measures, leading to inaccuracies in diagnosis. CNN models are a promising tool for faster diagnosis of malaria as the models can learn to classify blood smear images as parasitized and uninfected. This study aims to critically analyze different CNN models to classify blood smear images for faster diagnosis of Malaria.

1. Introduction

There is a conspicuous need and opportunity for AI and ML approaches to tend to medical needs Subsaharan Africa.(Mbunge & Batani, 2023). Malaria, a disease primarily caused by four Plasmodium parasites, continues to pose significant health risks globally, especially in sub-Saharan Africa where most cases occur. Despite a 45% reduction in malaria mortality rates from 2000 to 2014 due to improved prevention and control measures, the conventional microscopic diagnosis still faces challenges, such as low sensitivity and the lack of infrastructure in rural areas (Ragb et al., 2020). The integration of advanced technologies like deep convolutional neural networks (CNNs) offers promising solutions to these challenges. Recent studies have shown that using sophisticated CNN architectures or customized CNNs, not only outperforms traditional methods of diagnosis but also offers scalability and efficiency in automated malaria diagnosis. In this study, we have explored the approaches used by researchers globally for diagnosis of malaria by solving the binary classification problem of classifying images of RBCs obtained from blood smear images into Parasitized and Uninfected classes.

[*]Equal contribution [1]Department of Computer Science, Vanderbilt University, Nashville, TN, USA. Correspondence to: Arpit Ojha <arpit.ojha@vanderbilt.edu>, Shruti Jain <shruti.jain@vanderbilt.edu>.

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1.1. Data used

We have used the NIH Malaria Dataset (National Library of Medicine, n.d.) for this study. The dataset consists of blood smear slides based on blood samples taken from 150 infected and 50 healthy patients at the Chittatong Medical College Hospital in Bangladesh. Each image was then manually annotated as parasitized or uninfected at the Mahidol-Oxford Tropical Medicine Research Unit in Bangkok, Thailand. The blood smear slides contain various types of cells including white blood cells, platelets, and red blood cells but only the red blood cells need to be analyzed for diagnosis of malaria, so the blood smear images were segmented into images of individual red blood cells that are used for training. The dataset consists of labeled images of red blood cells. The dataset is publically available, You can find more information about the NIH malaria dataset at NIH malaria screener website.

1.2. Background

Diagnosis of malaria using Image processing and CNN is an ongoing research problem.(Mbunge & Batani, 2023). We reviewed several articles and have included only recent ones that are less than 5 years old in this review. Several articles have attempted and evaluated the performance of state-of-the-art deep neural networks for solving the approaches (Rahman et al., 2021), (Ragb et al., 2020), (Asiya et al., 2023), (Maqsood et al., 2021), (Sriporn et al., 2020), (Çinar & Yildirim, 2020), (Acherar et al., 2022). Our review shows that certain approaches have achieved very high accuracies upto 99% (Sriporn et al., 2020).

ML researchers have been tweaking complex models in different ways to increase accuracy. This section discusses such experiments. Some papers have also contributed towards data augmentation techniques that can be used for data preprocessing and other have also tested the model in real life.

Authors (Asiya et al., 2023), have critically analysed the performance of five different deep learning models in detecting malaria from these images: VGG-19, Inception-v3, ResNet-101, Inception Resnet-v2, and a custom CNN model. The results of the study indicated that Inception Resnet-v2 was the most accurate model, achieving an accuracy rate of 95.4%. This study is closest to our approach although our results differ from theirs, as can be seen in later sections.

(Ragb et al., 2020) By using a mixture of machine learning techniques such as transfer learning, a cyclical and constant learning rate, and ensemble methods, we have developed a model capable of accurately identifying parasitic cells within red blood smears.

(Sriporn et al., 2020) used Xception, with the state of the art activation function (Mish) and optimizer (Nadam), improved the effectiveness, as found by the outcomes of the convolutional neural model evaluation of these models for classification of the malaria disease from thin blood smear images. They could achieve a combined score of 99.28%.

(Minarno et al., 2023) used the Inception-V3 architecture to classify malaria Images. They achieved better results by changing the optimizer function to SGD optimizer, Adam optimizer and RMSprop optimizer. They concluded that the best results with 97% accuracy was achieved using the Inception-V3 model with the RMSprop optimizer.

Articles like (Çinar & Yildirim, 2020) emphasize the importance of image pre-processing. They have achieved results upto 97% accuracy by using Median and Gauss filters using DenseNet model. (Ragb et al., 2020) and others have also talked about data augmentation techniques such as rotations, flips, and color adjustments. We have adopted their technique in our work.

Some researchers have also used NIH malaria dataset for transfer learning. The robustness of deep learning models is often tested across various datasets to ensure their efficacy in real-world scenarios. For example, in (Rahman et al., 2021), they evaluated model performance on both proprietary and publicly available datasets, finding that models trained on well-curated data can generalize well and maintain high performance even on external datasets. They trained their CNN, VGG, Xception models on a different dataset and tested their model for transfer learning on the NIH Malaria dataset.

Authors like (Maqsood et al., 2021) have expanded the metrics of performance analysis by adding time complexity analysis for various CNN architectures.

Researchers like, (Acherar et al., 2022) have also performed patient-level validation to assess the generalizability of the CNN models in real-life conditions and yeilded promising results.

2. Methods

As described in section 1.2 the dataset is called NIH Malaria Dataset and was retrieved from NIH malaria screener website. The dataset consists of labeled images of red blood cells, that were segmented from images of blood smear slides of blood samples collected from several healthy individuals and malaria patients. As shown in Figure 1, the images of Red blood cells are categorized into two labels: Parasitized and Uninfected. The data set is balanced and consists of 13,780 samples of images for each of the two labels.

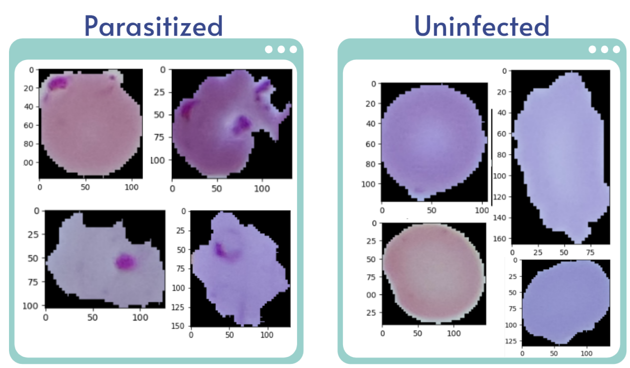

Figure 1. Sample images from the dataset. The dataset is categorized into two labels: Parasitized and Uninfected

2.1. Data Exploration

Image dimensions Checking Image dimensions is important before training CNN models. Table 1 summarizes the range of dimensions of these images and Figure 2 shows the distribution of the height and width. We calculated the average height and width of these images and resized all the images to these average dimensions before training the models. It was an important step to eliminate the possibility of significant image size differences between the two labels as we would have to do additional steps in the data pre-processing stage if the image sizes were drastically different.

Category	Height	Avg. H.	Width	Avg. W.
Parasitized	40 - 385	134.39	46 - 394	133.63
Uninfected	49 - 235	131.58	49 - 247	131.34

Table 1. Distribution of Height and Width for different labels in the image data

Histograms As can be seen in Figure 1, it is clear that

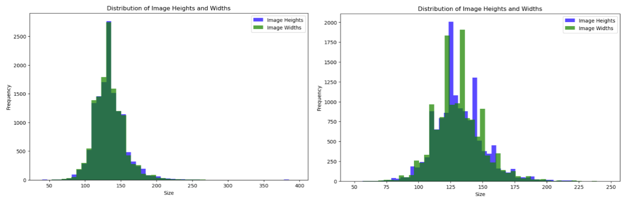

Figure 2. Distribution of image sizes for different labels; Parasitized(left) and Uninfected(right)

images of the Parasitized cells have more pixels of pink and purple hues, as also evident in the histograms presented in Figure 3. The histograms of parasitized images, on average, have more pixels of blue and red contributing to the pink-purple hues.

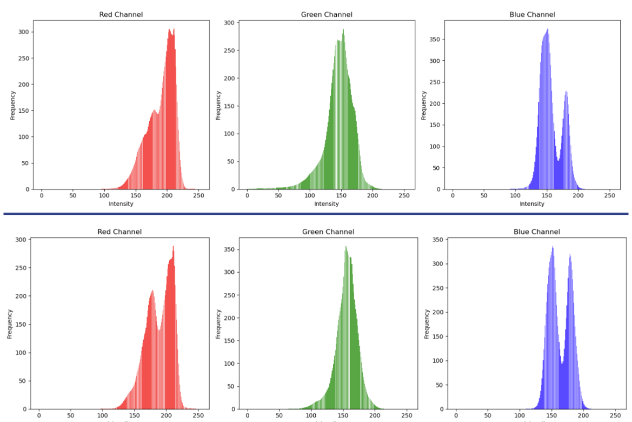

Figure 3. Average histograms of images (intensity v/s frequency of pixels); Parasitized(top) and Uninfected(bottom)

2.2. Data Pre-processing

Data Pre-processing comprised of two main steps: (i) Resizing and (ii) Data Augmentation. For resizing, all the images were resized to xx by xx, which was the average width and height of the images obtained in the data exploration phase.

Data Augmentation is one of the standard pre-processing techniques when using image data. In our literature review we observed that several papers have followed such procedures **cite**. This step is crucial to avoid overfitting when training large models. For the scope of this project, data Augmentation involves image transformation like rotation, width shift, and shear transformation as described below:

- **Rotation**: Randomly rotate the images by up to 20 degrees.
- **Width Shift**: Randomly shift the images horizontally by up to 10% of the total width.
- **Shear Transformation**: Randomly distort the shape of the image along one of its axes by up to 10%.

Data was then split 80-20% for training and validation respectively.

2.3. Architectures

Convolutional Neural Network (CNN) Models
The basic CNN models used in this project are variations of the standard CNN architecture. A typical CNN consists of an input layer, multiple convolutional layers with activation functions, pooling layers, and fully connected layers at the end. The convolutional layers learn features from the input data, the pooling layers reduce the spatial dimensions, and the fully connected layers perform the final classification or regression task.
The specific variations of CNN models used in this project include modifications to the number and configuration of the convolutional, pooling, and fully connected layers, as well as the choice of activation functions and regularization techniques. These variations are often designed to improve the model's performance on the specific problem at hand.

VGG 16
The VGG16 model is another influential deep convolutional neural network introduced by the Visual Geometry Group (VGG) at the University of Oxford (Simonyan & Zisserman, 2014). It features 16 layers that include 13 convolutional layers, 5 max-pooling layers, and 3 fully connected layers at the end of the network. Similar to VGG16, the convolutional layers predominantly use 3x3 filters, while the max-pooling layers utilize 2x2 filters with a stride of 2. VGG16 is well-regarded for its performance in various computer vision tasks such as image recognition and classification across a wide array of contexts. The architecture's depth, in conjunction with the consistent kernel size in convolutional layers, plays a crucial role in its ability to learn complex feature hierarchies.

Inception
The Inception model, also known as GoogLeNet, was developed by Google researchers (Szegedy et al., 2015). It is characterized by its "inception modules," which are complex building blocks that concatenate the outputs of multiple convolutional and pooling operations. This approach allows the model to capture features at different scales and resolutions, improving its performance on diverse tasks.
The Inception model has demonstrated strong performance in image classification, object detection, and other computer vision applications. Its modular design and efficient use of

computational resources have made it a popular choice for deployment in real-world systems.

Xception

The Xception model, short for "Extreme Inception," is a deep convolutional neural network proposed by researchers at Google (Chollet, 2017). It is an extension of the Inception model, where the standard Inception modules are replaced with depth-wise separable convolutions.

The Xception model has a simpler and more efficient architecture compared to the Inception model, while maintaining similar performance. It has shown promising results in tasks such as image classification, object detection, and semantic segmentation, making it a compelling choice for applications that require a balance between model complexity and performance.

2.4. Approach

As the literature we read pointed to these models, we implemented the models described below and evaluated them based on the performance metrics described in section 2.5. For a fair comparison between models, We ran all our experiments on a GPU with 32 GB RAM. Only the required Jupyter Kernel and the windows for model training were open at the time.

The details of experiments we conducted are as follows:

In the pursuit of creating an effective automated malaria detection system using deep learning, we have explored an array of Convolutional Neural Network (CNN) architectures. Our approach was twofold: firstly, we developed five custom CNN model variations designed specifically to process thin blood smear images and discern whether they are parasitized or uninfected. Secondly, we benchmarked these models against established architectures like Inception, VGG-16, and Xception to gain insights into the trade-offs between bespoke and state-of-the-art models in medical image classification.

Our initial variation provided a baseline model, consisting of three convolutional layers, each followed by a max-pooling layer to distill spatial features. To combat overfitting, dense layers with dropout components were incorporated. Building upon this, the second variation introduced a fourth convolutional layer to capture more complex patterns at the expense of an elevated parameter count. The third variation diverged from conventional activations by adopting leaky ReLU functions and global average pooling, which helped manage non-linearities and reduced the trainable parameter space without sacrificing the model's learning capability.

We continued our iterative model evolution with variations four and five, wherein batch normalization was integrated post-convolution to enhance learning stability and convergence. Variation five further streamlined the model by reducing the number of neurons in the dense layers following the flattening operation, a modification aimed at fine-tuning performance without overburdening computational resources.

In parallel with our tailored models, we extended our investigation to encompass Inception, VGG-16, and Xception given their distinct attributes and established performance. Inception's wider network with parallel kernel operations excels at capturing information across multiple scales. VGG-16's depth and smaller convolutional filters contribute to nuanced image analysis. Xception's efficiency is derived from its use of depthwise separable convolutions, which are adept at learning spatial hierarchies.

This multifaceted method reflects our dedication to uncovering an optimal CNN framework for medical diagnostics by balancing domain-specific customization with insights gleaned from industry-standard architectures. Our results are expected to shed light on how these various model configurations can be calibrated to enhance accuracy and reliability in the detection of malaria through blood smear analysis.

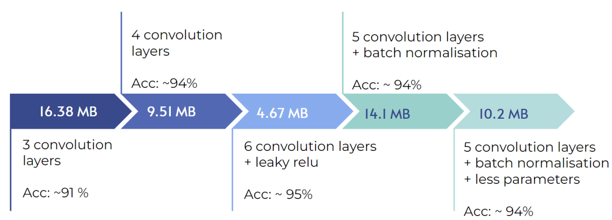

Figure 4. Experiments with the CNN models. The image highlights the variations of models and the parameters of in different models and the accuracy achieved.

2.5. Performance Metrics

In this project, we evaluated the performance of the various CNN models using the following key metrics:

Accuracy

Accuracy is the primary metric used to assess the overall performance of the CNN models. It represents the percentage of correct predictions made by the model. A higher accuracy value indicates that the model is better able to correctly classify the input samples.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Precision and Recall

Precision measures the proportion of true positive predictions out of the total positive predictions made by the model.

Recall (or sensitivity) measures the proportion of true positive predictions out of the total number of positive samples in the dataset. These metrics provide insights into the model's ability to correctly identify positive instances, which is particularly important for imbalanced datasets.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1-Score

The F1-score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. It ranges from 0 to 1, with 1 being the best score. The F1-score is useful for evaluating the overall effectiveness of the model, especially when dealing with imbalanced datasets.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Loss

The loss function, such as cross-entropy or mean squared error, was used to measure the difference between the model's predictions and the true labels during training and evaluation. Lower loss values indicate better model performance and more accurate predictions.

Model Size

Finally, we considered the size of the trained models, in terms of the number of parameters or the storage space required. This is an important factor for deployment on resource-constrained devices, where memory and computational resources may be limited.

Training Time

The training time, which measures the duration required to train the model on the given dataset, was also taken into account. This metric is important for understanding the computational resources and time investment needed to develop and optimize the CNN models.

By evaluating the CNN models using these comprehensive performance metrics, we were able to assess their effectiveness, robustness, and suitability for the classification task for malaria diagnosis.

3. Results

Our investigative journey in the realm of smart healthcare sought to enhance malaria diagnosis by leveraging the power of convolutional neural networks (CNNs). With a focus on optimizing accuracy and efficacy, we designed and experimented with five distinct CNN configurations, each uniquely tailored to the nuanced requirements of automated blood smear analysis.

CNN Variation 1 laid the foundational groundwork with a three-layer architecture. The simplicity of this model was its strength, achieving an admirable accuracy of around 90%, coupled with a precision of 0.9508, recall of 0.9029, and an F1 score of 0.9262, proving its potential for resource-constrained environments.

CNN Variation 2 expanded the depth with four convolutional layers and introduced TensorFlow sophistication, incorporating an early stopping mechanism. It demonstrated a consistent performance across various metrics with an accuracy of 94%, a precision, recall, and F1 score of 0.953, and only a slightly increased parameter size (9.51 MB) as compared to the baseline model.

CNN Variation 3 diverged by integrating six convolutional layers accompanied by leaky ReLU activation functions. Despite a modest accuracy of around 70%, it exhibited a remarkable recall of 0.9782 and an F1 score of 0.9582, emphasizing its prowess in identifying true positive cases—a critical aspect in medical diagnostics.

CNN Variation 4 appealed to both stability and efficiency by marrying a five-layer infrastructure with batch normalization techniques. It closely shadowed the peak, achieving nearly 95% accuracy and a compelling F1 score of 0.9529. Though its precision dipped to 0.9188, its consistent reliability marked it as an advancement in model design.

CNN Variation 5 was a refined iteration of the previous model, characterized by a reduced parameter footprint while maintaining comparative accuracy and F1 scores. This exemplified an efficient trade-off, suggesting that minimalism in design doesn't necessarily equate to reduced performance.

Against these bespoke variations, our study also recognized the contribution of more complex models—Xception, VGG-19, and Inception—with accuracy bands between 90% and 95%. These findings underlined the delicate balance between architectural intricacy and computational demands.

Our approach culminates in offering a spectrum of CNN architectures, each poised to cater to specific real-world application scenarios within smart healthcare. The results from our explorative analysis present evidence that while sophisticated models hold their merit, less complex architectures still stand as viable contenders, thriving under constrained operational environments. This research establishes a framework for future endeavors where computational efficiency and diagnostic accuracy must harmoniously coexist, driving forward the ambition of accessible and reliable healthcare diagnostics.

The CNN-3 model emerged as the best performer in this

	CNN-1	CNN-2	CNN-3	CNN-4	CNN-5	Xception	VGG-16	Inception
Accuracy	0.9174	0.9496	0.9534	0.9488	0.9485	0.95	0.93	0.91
Precision	0.95077	0.95325	0.939	0.91882	0.9243	0.98	0.97	0.94
Recall	0.9029	0.95325	0.97823	0.98965	0.985	0.92	0.87	0.88
f1-Score	0.92623	0.953247	0.95822	0.95292	0.9537	0.95	0.92	0.9
Loss	0.3025	0.1657	0.1403	0.1443	0.1516	0.158	0.24	0.2341
no. of parameters	16.38 MB	9.51 MB	4.67 MB	14.13 MB	10.02 MB	-	-	-
Training Time (s)	706	807	1672	1700	2430	9166	5661	1558

Table 2. Comparison of Results with different CNN Models

study, achieving an impressive accuracy of 95.34% and an F1-score of 0.9582, with a low loss of 0.1403. This model outperformed the other architectures tested, including the Xception model, which also showed strong results with an accuracy of 95% and an F1-score of 0.95. In contrast, the Inception model had the lowest accuracy among the tested models, at 91%. These results demonstrate the effectiveness of the CNN-3 architecture in classifying malaria-infected and uninfected red blood cell images, while the Inception model exhibited the weakest performance compared to the other models explored.

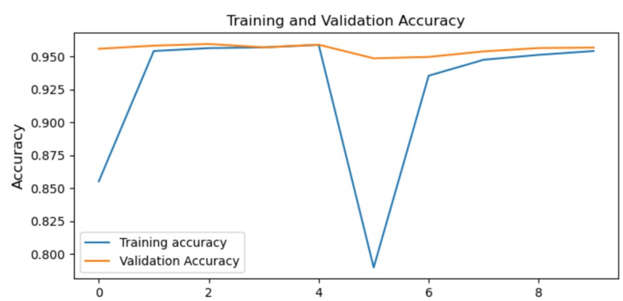

Figure 5. Best results achieved with the custom CNN model. The model Variation 3 with 6 convolution layers and leaky relu activation function, gave best results. The figure shows Training and Validation Accuracy of the model

4. Discussion and Conclusion

The convergence of artificial intelligence (AI) and healthcare, as explored in our study, promises to revolutionize the diagnostic landscape, bringing forth advancements that could notably improve patient outcomes. A case in point is the application of convolutional neural network (CNN) models for malaria detection, which aligns with the smart healthcare ideology in terms of integrating digital solutions into healthcare frameworks.

Our exploration into CNN architectures like VGG-16 and Inception ResNet-V2 has demonstrated their capability in achieving high accuracy in malaria detection. These findings

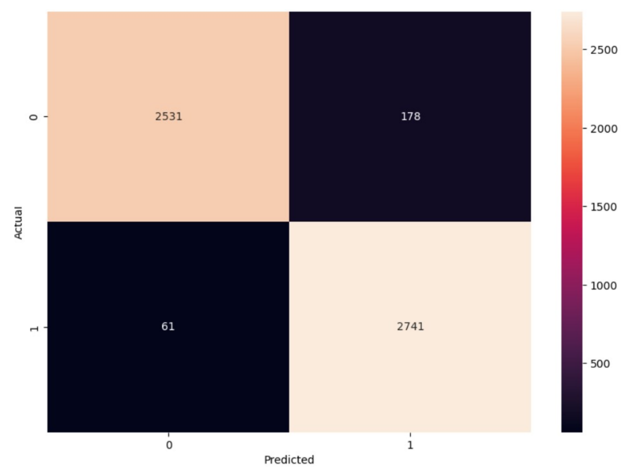

Figure 6. Best results achieved with the custom CNN model. The model Variation 3 with 6 convolution layers and leaky relu activation function, gave best results; The figure shows confusion matrix obtained with the validation set.

are indicative of the considerable promise that AI holds for automating and refining diagnosis processes. However, AI and ML implementations also raise ethical considerations, particularly in healthcare (Mbunge & Batani, 2023) Ensuring patient privacy, data security, and addressing potential biases within the training data are imperative concerns that must be recognized and acted upon to maintain trust and fairness in automated diagnostic systems. The AI systems must be transparent and explainable to facilitate clinical decision-making and to accommodate regulatory requirements.

Despite these advancements, this research has certain limitations which need to be considered. One of the constraints is the omission of inference time analysis for our models. The inference time is a critical factor in real-time or latency-sensitive scenarios where immediate response is necessary, such as in acute care settings or in the use of portable diagnostic devices. Additionally, the reliance on a single dataset source, the NIH Malaria Dataset, may affect the generalizability of the models. Although the dataset served as a sufficient basis for this specific study, the deployment of these models in varying locales with diverse malaria strains remains to be tested. Further research should consider the application of models across multi-source datasets to ensure robustness and adaptability to heterogeneous data.

In terms of future directions, it would be prudent to broaden the scope of our experimental framework to encompass an array of neural network architectures. Such exploration should involve a critical analysis of how altering model parameters affects performance, as well as an examination of the training procedures and the necessary computational resources for efficient model development and optimization. With regards to the practical deployment of AI in healthcare scenarios, a comprehensive assessment of scalability and real-world applicability is essential. This includes but is not limited to, the analysis of models in various settings, the impact of different strains of malaria across multiple regions, and the adaptation to datasets beyond the one utilized in this study.

In conclusion, this research, together with insights from smart healthcare trends, solidifies the stance that ML and AI have transformative potential in medical diagnostics. While ethical considerations remain an integral part of AI deployment in healthcare, the current investigation and subsequent analysis provide foundational evidence for the effective application of CNNs in malaria diagnosis. Moving forward, addressing the methodological limitations and ethical concerns, and diversifying the data sources will be instrumental in advancing smart healthcare solutions to their full potential.

5. Acknowledgments

We would like to express our sincere gratitude to our professor, Dr. Jonathan Sprinkle, for their invaluable guidance, expertise, and encouragement throughout this project. We are also thankful to our teaching assistant, Kuan-I Chung (Gary), for their valuable feedback and support during the course of this work. We are specially grateful for the support we got while we were trying to pivot our project. The constructive feedback we got from both, the professor of the course and the TA was invaluable to this study.

Additionally, we would like to acknowledge the Institute for Software Integrated Systems at the Vanderbilt University for providing us with the resources and infrastructure necessary to conduct this research effectively.

We have referred to publically avaialble code from Kaggle and GitHub and used ChatGPT and stackoverflow for resolving errors. Not all functions were developed from scratch, we have used functions from python libraries like scikit-learn and Keras for this project.

Both our teammates have contributed equally to the project, from ideation to development.

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A. Code

The submitted zipfile also contains the Jupiter notebooks (`*.ipynb`) that were used for the project.

`cnn-Variation1.ipynb` script executes preprocessing, data augmentation, and develops a three-layer CNN, monitoring model performance through accuracy and loss.

`cnn-Variation2.ipynb` employs TensorFlow to craft and assess a four-layer CNN model, utilizing early stopping and comprehensive performance metrics on cell images.

`cnn-Variation3.ipynb` describes the creation of a six-layer CNN with leaky ReLU, evaluated for its ability to classify images based on accuracy and F1-score.

`cnn-Variation4.ipynb` refines a five-layer CNN with batch normalization for cell image classification, with extensive performance visualizations.

`cnn-Variation5.ipynb` enhances a CNN with five convolution layers, batch normalization, and presents the model's efficacy via performance metrics and visual analysis using low dropout rate.

`VGG and Inception.ipynb` manages malaria image classification tasks with VGG16 and InceptionV3 architectures, focuses on model accuracy, and includes model preservation steps.

`Xception.ipynb` builds a TensorFlow and Keras-based model using the Xception architecture, progresses through augmented image training with early stopping, and depicts the model's performance with metrics and Matplotlib.

`exploratory analysis.ipynb` file was used in the beginning for exploring the dataset.